

C5.2 Controlling reactions

Summary

The rate and yield of a chemical reaction can be altered by changing the physical conditions.

Underlying knowledge and understanding

Learners should be familiar with the action of catalysts in terms of rate of reaction. They should know the term surface area and what it means.

Common misconceptions

Learners often misinterpret rate graphs and think that catalysts take part in reactions and run out/get used up.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.
All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
CM5.2i	arithmetic computation, ratio when measuring rates of reaction	M1a, M1c
CM5.2ii	drawing and interpreting appropriate graphs from data to determine rate of reaction	M4b, M4c
CM5.2iii	determining gradients of graphs as a measure of rate of change to determine rate	M4d, M4e
CM5.2iv	proportionality when comparing factors affecting rate of reaction	M1c

Topic content		Opportunities to cover:		Practical suggestions
Learning outcomes	To include	Maths	Working scientifically	
C5.2a suggest practical methods for determining the rate of a given reaction		M1a, M1c	WS1.2b, WS1.2c, WS1.2d, WS2a, WS2b	Rate of reaction experiments. (PAG C1, PAG C8) Disappearing cross experiment. (PAG C8) Magnesium and acid, marble chip and acid. (PAG C1, PAG C8)
C5.2b interpret rate of reaction graphs	1/t is proportional to rate and gradients of graphs (not order of reaction)	M4b, M4c	WS1.3a, WS1.3b, WS1.3c, WS1.3d, WS1.3e, WS1.3f, WS1.3g, WS1.3h, WS1.3i, WS2b	Marble chip and acid or magnesium and acid experiments either measuring reaction time or the volume of gas over time. (PAG C1, PAG C7, PAG C8)
C5.2c describe the effect of changes in temperature, concentration, pressure, and surface area on rate of reaction		M4d, M4e	WS1.4c	Varying surface area with marble chips and hydrochloric acid. (PAG C1, PAG C8)
C5.2d explain the effects on rates of reaction of changes in temperature, concentration and pressure in terms of frequency and energy of collision between particles			WS1.4c	Reaction of magnesium and acid with different temperatures of acid – measure reaction times. (PAG C1, PAG C8)
C5.2e explain the effects on rates of reaction of changes in the size of the pieces of a reacting solid in terms of surface area to volume ratio		M1c		

C5.2f	describe the characteristics of catalysts and their effect on rates of reaction				
C5.2g	identify catalysts in reactions			WS1.4a	Catalysis of hydrogen peroxide with various black powders including MnO_2. (PAG C1, PAG C8) Catalysis of reaction of zinc with sulfuric acid using copper powder. (PAG C1, PAG C8)
C5.2h	explain catalytic action in terms of activation energy	reaction profiles			
C5.2i	recall that enzymes act as catalysts in biological systems				